



# IUPAC NOMENCLATURE & STRUCTURAL ISOMERISM

## 1. SECTION (A) : FUNDAMENTAL OF ORGANIC CHEMISTRY

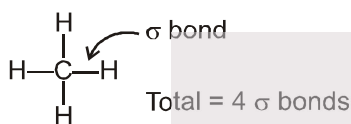
Th1:

### 1.1 Bonding in organic compounds :

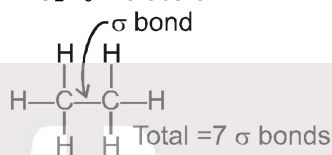
Two types of covalent bond exist in organic compounds.

(a) **Sigma bond ( $\sigma$ )** : The covalent bond formed between 2 atoms by mutual sharing of 1 pair of  $e^-$ . It is denoted by  $(-)$ .

Ex. In  $\text{CH}_4$  molecule



In  $\text{C}_2\text{H}_6$  molecule



(b) **Multiple bond ( $\pi$ )** :

Any other bond with  $\sigma$  bond is  $\pi$  bond.

Ex.

|                           |                    |                            |                    |
|---------------------------|--------------------|----------------------------|--------------------|
| (i)<br>In ethane Molecule |                    | (ii)<br>In ethyne molecule |                    |
|                           | Total $\sigma = 5$ |                            | Total $\sigma = 3$ |
|                           | $\pi = 1$          |                            | $\pi = 2$          |

Que. Calculate  $\sigma$  and  $\pi$  bond in following compounds.

(a)  $\text{HC} \equiv \text{CCH}=\text{CHCH}_3$

(b)  $\text{CH}_2 = \text{C} = \text{CHCH}_3$

Sol.

(a)  $\sigma_{\text{C-C}} : 4$  ;  $\sigma_{\text{C-H}} : 6$  ;  $\pi_{\text{C=C}} : 1$  ;  $\pi_{\text{C}\equiv\text{C}} : 2$

(b)  $\sigma_{\text{C-C}} : 3$  ;  $\sigma_{\text{C-H}} : 6$  ;  $\pi_{\text{C=C}} : 2$

### 1.2 Some important definitions :

D1:

- (i) **Catenation** : The property of atoms of an element to link with one another forming chains of identical atoms is called catenation.

D2:

- (ii) **Homologous series**: Homologous series may be defined as a series of similarly constituted compounds in which the members possess the same functional group, have similar chemical characteristics and have a regular gradation in their physical properties. The two consecutive members differ in their molecular formula by  $\text{CH}_2$ .



## Th2:

## 1.3 Structural representation of organic compounds :

There are three ways for representation of organic compounds :

## (i) Complete structural formula:

Such a structural formula focuses on the electrons involved in bond formation. A single dash (–) represents a single bond, double dash (=) is used for double bond and a triple dash ( $\equiv$ ) represents triple bond. Lone- pairs of electrons on heteroatoms (e.g., oxygen, nitrogen, sulphur, halogens etc.) may or may not be shown.

## (ii) Condensed structural formula:

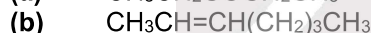
Structural formulas can be further abbreviated by omitting some or all of the dashes representing covalent bonds and by indicating the number of identical groups attached to an atom by a subscript. The resulting expression of the compound is called a condensed structural formula.

## (iii) Bond line formula:

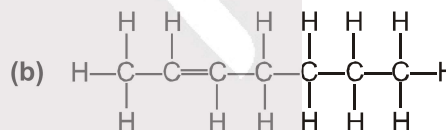
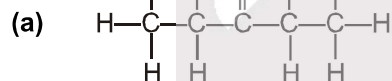
In this formula, carbon and hydrogen atoms are not shown and the lines representing carbon-carbon bonds are drawn in a zig-zag fashion. The only atoms specifically written are oxygen, chlorine, nitrogen etc.

| Condensed form      | Expanded form | Bond line form |
|---------------------|---------------|----------------|
| $C(CH_3)_4$         |               |                |
| $CH_3(CH_2)_2CH_3$  |               |                |
| $H_2N(CH_2)_2OCH_3$ |               |                |

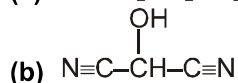
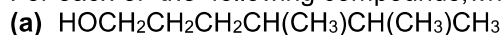
**Que.** Expand each of the following condensed formulas into their complete structural formulas.



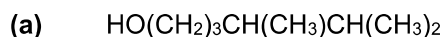
**Sol.**



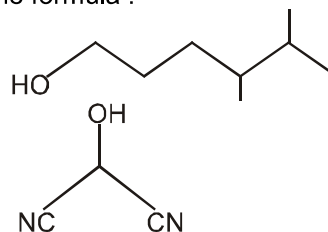
**Que.** For each of the following compounds, write a condensed formula and also their bond-line formula.



**Sol.** Condensed formula :



Bond-line formula :

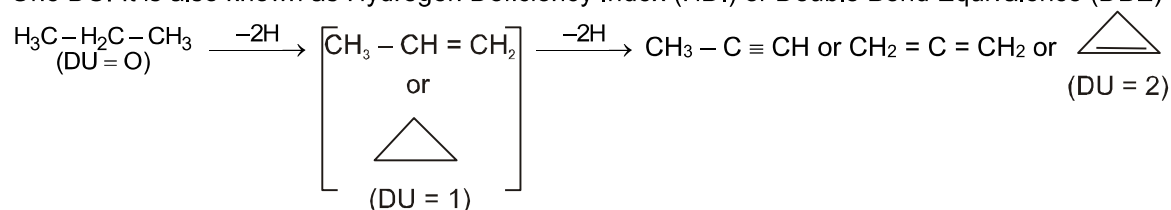


**Th3:****1.4 Degree of Unsaturation (DU) :**

The presence of double bonds or rings within a molecule is indicated by a quantity called degree of unsaturation.

**Applications:** To identify the no. of  $\pi$  bonds or rings and also helpful in determining the structure of the molecule.

**D3: Definition:** Deficiency of 2H atoms with respect to fully saturated acyclic hydrocarbon is equivalent to One DU. It is also known as Hydrogen Deficiency Index (HDI) or Double Bond Equivalence (DBE)



$$\text{Degree of unsaturation (D.U.)} = \frac{(2n + 2) - (\text{No. of H atoms} + \text{No. of X atoms} - \text{No. of N atoms})}{2}$$

Where n = number of carbon atoms in the molecule

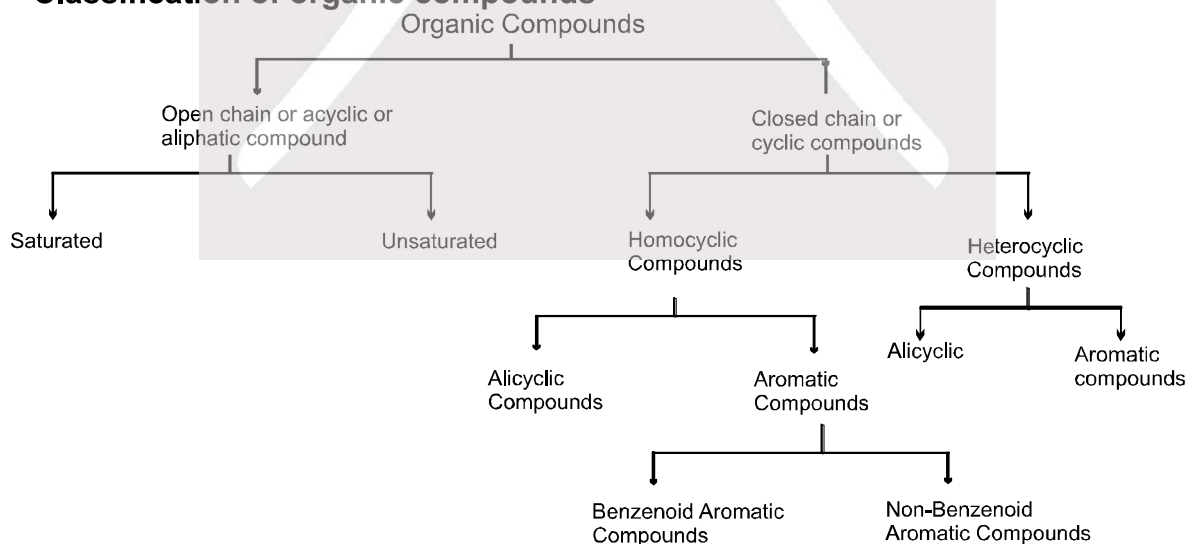
**Note :** Total no. of cyclic rings + double bonds will gives us degree of unsaturation.

One double bond = one DU

One ring = one DU

One triple bond = two DU

|     |       |                             |                                                        |      |  |        |
|-----|-------|-----------------------------|--------------------------------------------------------|------|--|--------|
| Ex. | (i)   | $\text{CH}_2 = \text{CH}_2$ | $\text{DU} = \frac{(2 \times 2 + 2) - 4}{2} = 2/2 = 1$ | (ii) |  | DU = 2 |
|     | (iii) |                             | DU = 4                                                 | (iv) |  | DU = 7 |

**Th4:****1.5 Classification of organic compounds**



## Th5:

## 1.6 Organic compounds and functional groups

Number of known organic compounds is much more than inorganic compounds but, it has been possible to group them into classes or families based on their structural features. This has given organic chemistry a logical and systematic shape. Examples are as follows :

1.6.1 Alkanes [general formula  $C_nH_{2n+2}$  where  $n = 1, 2, 3, \dots$ ]

These are open-chain aliphatic saturated hydrocarbon which have no functional groups. These are also called **paraffins**.

|                                          |   |         |                                       |   |        |
|------------------------------------------|---|---------|---------------------------------------|---|--------|
| $n = 1 \Rightarrow CH_4$                 | – | Methane | $n = 2 \Rightarrow C_2H_6$            | – | Ethane |
| $n = 3 \Rightarrow CH_3CH_2CH_3$         | – | Propane | $n = 4 \Rightarrow CH_3CH_2CH_2CH_3$  | – | Butane |
| $n = 5 \Rightarrow CH_3CH_2CH_2CH_2CH_3$ | – | Pentane | $n = 10 \Rightarrow CH_3(CH_2)_8CH_3$ | – | Decane |

1.6.2 Alkenes [general formula  $C_nH_{2n}$  where  $n = 2, 3, \dots$ ]

Alkenes are open chain unsaturated hydrocarbons and having carbon-carbon double bonds ( $C=C$ ). These are also called **alkylenes or olefins**. The first three members are generally named by their common names.

|     |                         |                             |                                 |                                                           |
|-----|-------------------------|-----------------------------|---------------------------------|-----------------------------------------------------------|
| Ex. | $CH_2=CH_2$<br>ethylene | $CH_3-CH=CH_2$<br>propylene | $CH_3-CH_2-CH=CH_2$<br>butylene | $CH_3-\overset{\overset{CH_3}{ }}{C}=CH_2$<br>Isobutylene |
|-----|-------------------------|-----------------------------|---------------------------------|-----------------------------------------------------------|

1.6.3 Alkynes [general formula  $C_nH_{2n-2}$  where  $n = 2, 3, \dots$ ]

Unsaturated aliphatic hydrocarbons containing a carbon-carbon triple bond are called alkynes.

The common names of a few simple alkynes are given below.

|                              |   |                            |
|------------------------------|---|----------------------------|
| $CH \equiv CH$               | – | Acetylene                  |
| $CH_3-C \equiv CH$           | – | Methyl acetylene           |
| $CH_3-CH_2-C \equiv CH$      | – | Ethylacetylene             |
| $CH_3-C \equiv C-CH(CH_3)_2$ | – | Methyl isopropyl acetylene |

## 1.6.4 Some names of hydrocarbon groups

## (A) Alkyl, Alkenyl &amp; Alkynyl groups

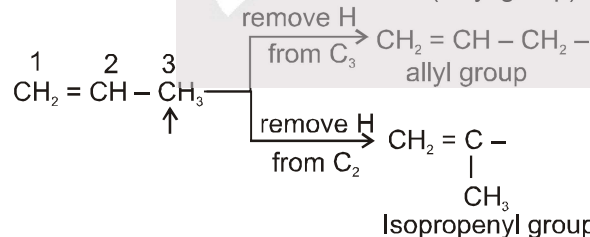
Alkane ( $C_nH_{2n+2}$ )  $\xrightarrow{-H}$  Alk + yl ( $C_nH_{2n+1}$ )

Alkene ( $C_nH_{2n}$ )  $\xrightarrow{-H}$  Alken + yl ( $C_nH_{2n-1}$ )

Alkyne ( $C_nH_{2n-2}$ )  $\xrightarrow{-H}$  Alkyn + yl ( $C_nH_{2n-3}$ )

|     |                                                                                |
|-----|--------------------------------------------------------------------------------|
| Ex. | methane $\xrightarrow[-yl]{-ane}$ methyl ( $CH_4 \xrightarrow{-H} -CH_3$ )     |
|     | propane $\xrightarrow[-yl]{-ane}$ propyl ( $C_3H_8 \xrightarrow{-H} -C_3H_7$ ) |

$CH_2 = CH_2 \xrightarrow{\text{remove H}} -CH=CH_2$  (vinyl group)/ethenyl.

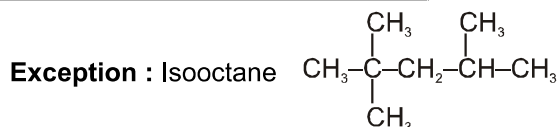


$HC \equiv CH \xrightarrow{-H} HC \equiv C -$  (Ethynyl);  $H_3C-C \equiv CH \xrightarrow{-H} H_3C-C \equiv C -$  (propynyl)

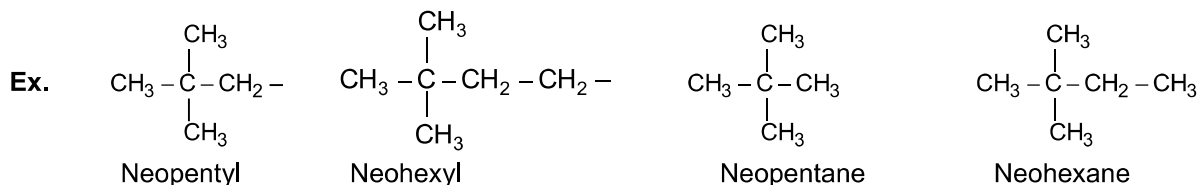
(B) Iso alkyl group : A compound having  $-\overset{\overset{CH_3}{|}}{CH}-CH_3$  group is called iso alkyl group.

|     |                                           |                                                |                                                     |                                                         |
|-----|-------------------------------------------|------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Ex. | $CH_3-\underset{\underset{CH_3}{ }}{CH}-$ | $CH_3-\underset{\underset{CH_3}{ }}{CH}-CH_2-$ | $CH_3-\underset{\underset{CH_3}{ }}{CH}-CH_2-CH_2-$ | $CH_3-\underset{\underset{CH_3}{ }}{CH}-CH_2-CH_2-CH_3$ |
|     | Iso propyl                                | Iso butyl                                      | isopentyl                                           | Iso pentane                                             |





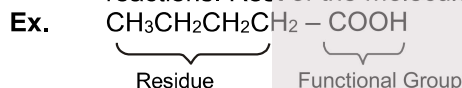
(C) **Neo alkyl group** : Compound having  $\left( \text{CH}_3 - \overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}} - \text{CH}_2 - \right)$  group is called neo alkyl group.



### 1.6.5 Functional group and residue

The characteristic group of atom which decide the physical and chemical properties of an organic molecule is called functional group.

Functional group is that portion of molecule which is highly reactive and takes part in chemical reactions. Rest of the molecule is called Residue.



## 2. SECTION (B) : IUPAC-NOMENCLATURE OF ALKANE & CYCLOALKANE

Th6:

### 2.1 IUPAC system of nomenclature

The IUPAC name of any organic compound consists of maximum five parts in the following sequence.

**Secondary prefix + Primary prefix + Word root + Primary suffix + Secondary suffix**

### 2.2 Word root :

It is the basic unit of the name. It denotes the number of carbon atoms present in the principal chain (the longest possible continuous chain of carbon atoms including the functional group and based upon the common names of alkanes) of the organic molecules.

| No. of carbon atoms in parent chain | Word root (Alk) | No. of carbon atoms in parent chain | Word root (Alk) | No. of carbon atoms in parent chain | Word root (Alk) |
|-------------------------------------|-----------------|-------------------------------------|-----------------|-------------------------------------|-----------------|
| 1                                   | Meth            | 9                                   | Non             | 20                                  | Icos            |
| 2                                   | Eth             | 10                                  | Dec             | 30                                  | Triacont        |
| 3                                   | Prop            | 11                                  | Undec           | 40                                  | Tetracont       |
| 4                                   | But             | 12                                  | Dodec           | 50                                  | Pentacont       |
| 5                                   | Pent            | 13                                  | Tridec          | 60                                  | Hexacont        |
| 6                                   | Hex             | 14                                  | Tetradec        | 70                                  | Heptacont       |
| 7                                   | Hept            | 15                                  | Pentadec        | 80                                  | Octacont        |
| 8                                   | Oct             | 16                                  | Hexadeca        | 100                                 | Cent & Hect     |

### 2.3 Primary suffix.

A primary suffix is always added to the word root to indicate whether the carbon chain is saturated or unsaturated. The three basic primary suffixes are given below :

| Type of carbon chain                 | Primary suffix | General name |
|--------------------------------------|----------------|--------------|
| (a) Saturated                        | – ane          | Alkane       |
| (b) Unsaturated with one double bond | – ene          | Alkene       |
| (c) Unsaturated with one triple bond | – yne          | Alkyne       |



If the parent carbon chain contains two, three or more double or triple bonds, numerical prefix such as di (for two), tri (for three), tetra (for four) etc. are added to the primary suffix. For example,

| Type of carbon chain                  | Primary suffix | General name |
|---------------------------------------|----------------|--------------|
| (a) Unsaturated with two double bonds | (a) + diene    | Alkadiene    |
| (b) Unsaturated with two triple bonds | (a) + diyne    | Alkadiyne    |
| (c) Both double and triple bonds      | enynes         | Alkenyne     |

## 2.4 Secondary suffix :

A secondary suffix is then added to the primary suffix to indicate the nature of the functional group present in the organic compounds. Secondary suffix of important functional groups are given below in their decreasing order of seniority.

|     | Class                                                                                                                                                   | Name                  | Suffix                                 | Prefix                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------|---------------------------------|
| 1.  | R – COOH                                                                                                                                                | Alkanoic Acid         | – oic acid (carboxylic acid)           | Carboxy                         |
| 2.  | R – SO <sub>3</sub> H                                                                                                                                   | Alkane sulphonic Acid | – sulphonic acid                       | sulpho                          |
| 3.  | $\begin{array}{c} \text{R} - \text{C} - \text{O} - \text{C} - \text{R} \\ \parallel \quad \quad \parallel \\ \text{O} \quad \quad \text{O} \end{array}$ | Alkanonic Anhydride   | – oic anhydride (carboxylic anhydride) | -----                           |
| 4.  | R – COOR                                                                                                                                                | Alkyl alkanoate       | – oate (carboxylate)                   | alkoxy carbonyl or alkanoyl oxy |
| 5.  | $\begin{array}{c} \text{R} - \text{C} - \text{X} \\ \parallel \\ \text{O} \end{array}$                                                                  | Alkanoyl halide       | – oyl halide (carbonyl halide)         | halo carbonyl                   |
| 6.  | $\begin{array}{c} \text{R} - \text{C} - \text{NH}_2 \\ \parallel \\ \text{O} \end{array}$                                                               | Alkanamide            | – amide (carboxamide)                  | carbamoyl                       |
| 7.  | R – C ≡ N                                                                                                                                               | Alkanenitrile         | – nitrile (carbonitrile)               | cyano                           |
| 8.  | $\begin{array}{c} \text{R} - \text{C} - \text{H} \\ \parallel \\ \text{O} \end{array}$                                                                  | Alkanal               | – al (carbaldehyde)                    | formyl / oxo                    |
| 9.  | $\begin{array}{c} \text{R} - \text{C} - \text{R} \\ \parallel \\ \text{O} \end{array}$                                                                  | Alkanone              | – one                                  | oxo                             |
| 10. | R – OH                                                                                                                                                  | Alkanol               | – ol                                   | hydroxy                         |
| 11. | R – SH                                                                                                                                                  | Alkanethiol           | – thiol                                | mercapto                        |
| 12. | R – NH <sub>2</sub>                                                                                                                                     | Alkanamine            | – amine                                | amino                           |

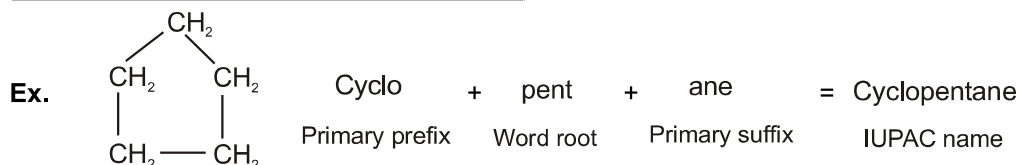
The following examples illustrate the use of word root, primary suffix and secondary suffix in naming of organic compounds.

| Organic Compounds                                               | Word root | Primary suffix | Secondary suffix | IUPAC name     |
|-----------------------------------------------------------------|-----------|----------------|------------------|----------------|
| CH <sub>3</sub> CH <sub>2</sub> OH                              | Eth       | an(e)          | ol               | Ethanol        |
| CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> NH <sub>2</sub> | Prop      | an(e)          | amine            | Propanamine    |
| CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> COOH            | But       | an(e)          | oic acid         | Butanoic acid  |
| CH <sub>3</sub> CH <sub>2</sub> CN                              | Prop      | an(e)          | nitrile          | Propanenitrile |
| CH <sub>2</sub> =CHCHO                                          | Prop      | en(e)          | al               | Propenal       |
| HC ≡ CCOOH                                                      | Prop      | yn(e)          | oic acid         | Propynoic acid |

## 2.5 Primary prefix :

A primary prefix is used simply to distinguish cyclic from acyclic compounds.

For example, in case of carbocyclic compounds, (cyclic compounds containing only carbon atoms in the ring), a primary prefix, **cyclo** is used immediately before the word root. Thus,



If the prefix cyclo is not used, it simply indicates that the compound is acyclic or open chain.

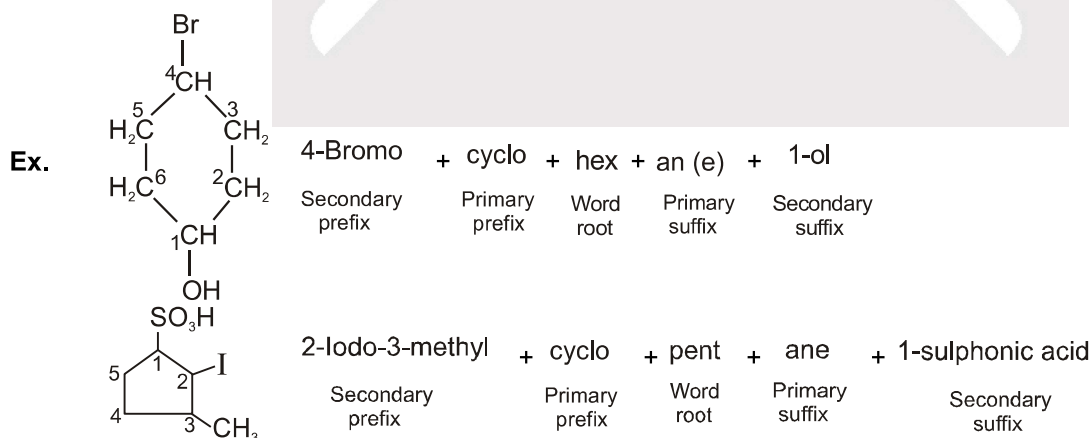
## 2.6 Secondary prefix :

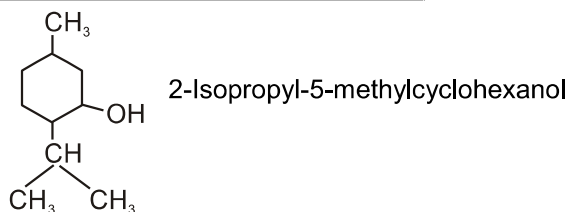
In IUPAC system of nomenclature, certain groups are not considered as functional groups but are treated as substituents. These are called secondary prefixes and are added immediately before the word root (or the primary prefix in case of carbocyclic compounds) in alphabetical order to denote the side chains or substituent groups. The secondary prefixes for some groups which are always treated as substituent groups (regardless of the fact whether the organic compound is monofunctional or polyfunctional) are given below :

| Substituent group                                   | Secondary prefix | Substituent group                                        | Secondary prefix |
|-----------------------------------------------------|------------------|----------------------------------------------------------|------------------|
| – F                                                 | Fluoro           | – OCH <sub>3</sub> (–OMe)                                | Methoxy          |
| – Cl                                                | Chloro           | – OC <sub>2</sub> H <sub>5</sub> (–OEt)                  | Ethoxy           |
| – Br                                                | Bromo            | – R                                                      | Alkyl            |
| – I                                                 | Iodo             | – CH <sub>3</sub> (–Me)                                  | Methyl           |
| – NO <sub>2</sub>                                   | Nitro            | – C <sub>2</sub> H <sub>5</sub> (–Et)                    | Ethyl            |
| – NO                                                | Nitroso          | – CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub> (n-Pr) | n-Propyl         |
| $\text{--}\overset{\oplus}{\text{N}}\equiv\text{N}$ | Diazo            | – CH(CH <sub>3</sub> ) <sub>2</sub> (–iPr)               | Isopropyl        |
| – OR                                                | Alkoxy           | – C(CH <sub>3</sub> ) <sub>3</sub> (t-Bu)                | t-Butyl          |

Example :

| Organic compounds                                              | Secondary prefix | Word root | Primary suffix | IUPAC name   |
|----------------------------------------------------------------|------------------|-----------|----------------|--------------|
| CH <sub>3</sub> CH <sub>2</sub> – Br                           | Bromo            | eth       | ane            | Bromoethane  |
| CH <sub>3</sub> – NO <sub>2</sub>                              | Nitro            | meth      | ane            | Nitromethane |
| C <sub>2</sub> H <sub>5</sub> – OC <sub>2</sub> H <sub>5</sub> | Ethoxy           | eth       | ane            | Ethoxyethane |





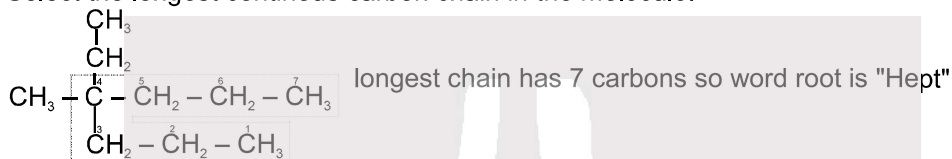
|      |                  |   |                      |
|------|------------------|---|----------------------|
| Here | Secondary prefix | = | 2-Isopropyl-5-methyl |
|      | Primary prefix   | = | cyclo                |
|      | Word root        | = | hex                  |
|      | Primary suffix   | = | an(e)                |
|      | Secondary suffix | = | ol                   |

Th7:

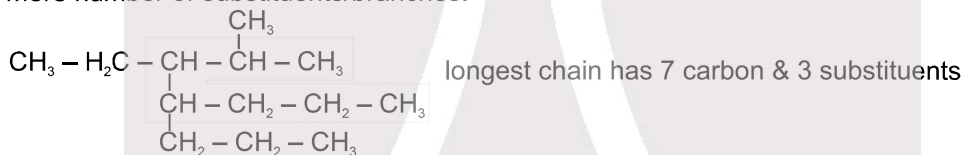
### 3. IUPAC NOMENCLATURE OF BRANCHED / COMPLEX ALKANES

#### 3.1 Parent carbon chain selection :

(a) Select the longest continous carbon chain in the molecule.

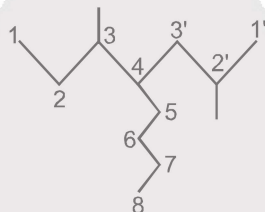


(b) When chains of equal lengths are competing for selection then that chain is selected which has more number of substituents/branches.



(c) When the number of substituents are same then the substituents at the nearest positions from the either end is prefer for parent chain selection.

Ex. Here, 2 choices for longest chain



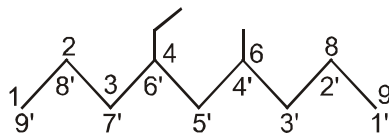
Chain- (A) 1-2-3-4-5-6-7-8

Chain- (B) 1'-2'-3'-4'-5'-6'-7'-8'

Chain- (A) & Chain- (B) both have 2 substituents but in chain-B substituent is nearer (at 2<sup>nd</sup> position) than in chain-A (at 3<sup>rd</sup> position). So, chain-B will be preferred.

(d) If the two substituents are found in equivalent positions the lower number is given to the one coming first in the alphabetical order.

Ex. Here, 2 choices for longest chain



Chain- (A) 1-2-3-4-5-6-7-8-9

Chain- (B) 1'-2'-3'-4'-5'-6'-7'-8'-9'

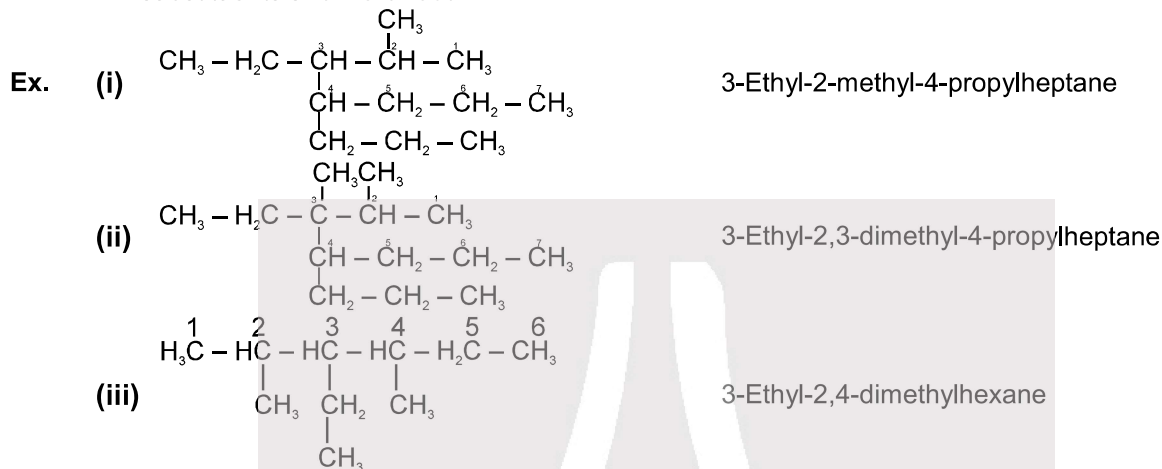
In both chain-A & chain-B, substituents are at same position (4<sup>th</sup>). In chain-A substituent is ethyl & in chain-B, it is methyl. Alphabetically ethyl will be preferred. So, chain-A is selected.

#### 3.2 Numbering of the parent carbon chain :

The numbering is done in such a way that the branched carbon atoms get the lowest possible number :

**Note :**

- (1) Write the substituents in place of secondary prefix with their appropriate locations in alphabetical order.
- (2) If the same substituent occurs more than once in the molecule, the prefix di (for two), tri (for three), etc. are used to indicate how many times it appears.
- (3) Prefixes di, tri, tetra etc. are not considered in deciding alphabetical order for simple substituents but considered for complex substituents.
- (4) Iso & Neo is considered for alphabetical seniority order.
- (5) Numbers are separated from each other by commas(,).
- (6) Numbers are separated from words by hyphens and there is no break between name of substituents and word root.

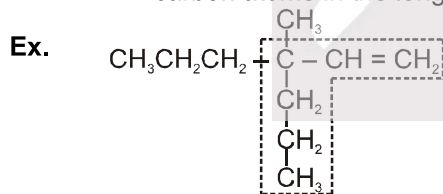


## 4. SECTION (C) : IUPAC-NOMENCLATURE OF ALKENE, CYCLOALKENE, POLYENES & ALKYNE

**Th8:****4.1 IUPAC nomenclature of Alkenes/Alkynes/Alkenyne****4.1.1 Alkenes :**

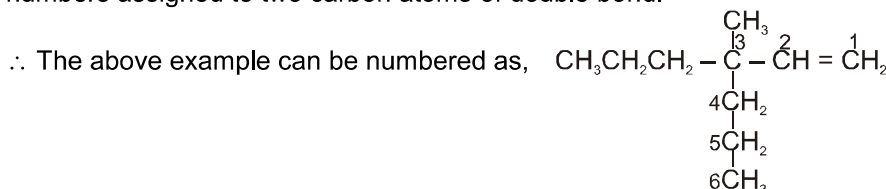
Functional group :  $\text{—C=C—}$

- (1) Select the longest carbon chain containing carbon-carbon double bond. This need not be the longest chain in the compound as a whole. Parent name will be alkene corresponding to number of carbon atoms in the longest chain.

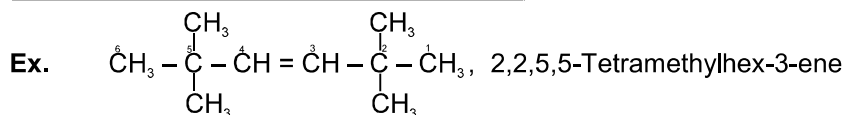


Longest chain has 6 atoms  $\Rightarrow$  parent name = hexane

- (2) Carbon atoms in the longest chain is numbered from that end in such a way that doubly bonded carbon atom gets the lowest number. The position of double bond is indicated by the smaller of the numbers assigned to two carbon atoms of double bond.

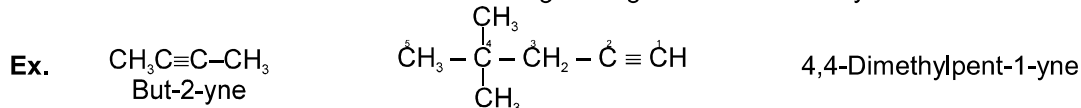


Position of double bond will be indicated as no. 1, Hence name will be 3-Methyl-3-propylhex-1-ene



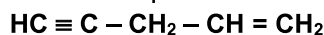
#### 4.1.2 Alkynes

Parent chain selection and numbering of longest chain is exactly same as that for alkenes.



#### 4.1.3 Alkenyne (containing both double and triple bonds)

Numbering is done in a manner that double and triple bonds get the lowest possible number. If double bond and triple bond both have same number then double bond is prefer over triple bond.



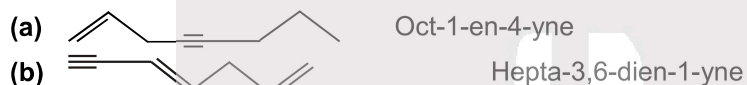
1 2 3 4 5 (numbering is done from alkyne)

(wrong)

5 4 3 2 1 (numbering is done from alkene)

(Correct)

Ex.



Th9:

## 5. IUPAC NOMENCLATURE OF ALICYCLIC COMPOUNDS

(1) The names of alicyclic compounds are obtained by adding the prefix "cyclo"

Ex.



Cyclobutane

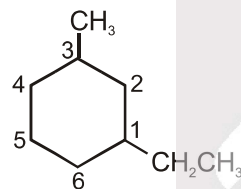


Cyclopentene

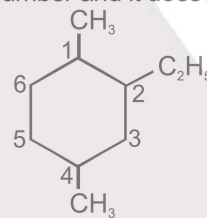
(2)

The numbering of the carbon atoms in the ring is done in such a way that the substituent which comes first in the alphabetical order is given the lowest possible number and it does not violate the lowest set of locants rule.

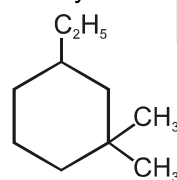
Ex.



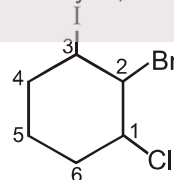
1-Ethyl-3-methyl cyclohexane



2-Ethyl-1,4-dimethyl cyclohexane



3-Ethyl-1,1-dimethyl cyclohexane

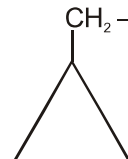


2-Bromo-1-chloro-3-iodocyclohexane

(3)

When the ring contains more or equal number of carbon atoms than the alkyl group attached to it, then it is named as a derivative of cycloalkane and the alkyl group is treated as substituent

Ex.



Propylcyclopropane

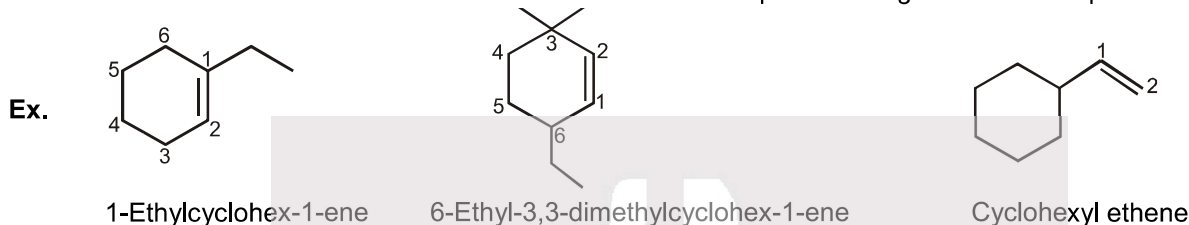




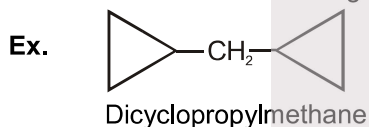
- (4) The alkane chain contains greater number of carbon atoms than present in the ring, then the compound is considered as the derivative of alkane and the ring is designated as substituent.



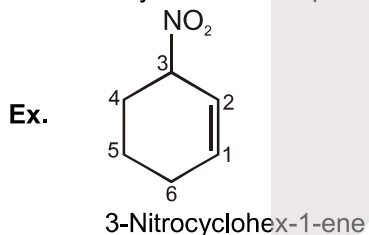
- (5) If ring has unsaturation and side chain is saturated then ring is selected as parent chain.  
 If side chain has unsaturation and ring is saturated then side chain is selected as parent chain.  
 If both have unsaturation the chain with maximum unsaturation has selected as parent chain.  
 If equal unsaturation then longest chain is selected as parent chain.  
 If unsaturation and number of carbon atoms both are equal then ring is selected as parent chain.



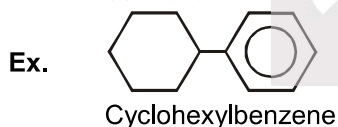
- (6) If more than one alicyclic ring is attached to a single chain then the compound is named as a derivative of alkane and the ring are treated as a substituent group.



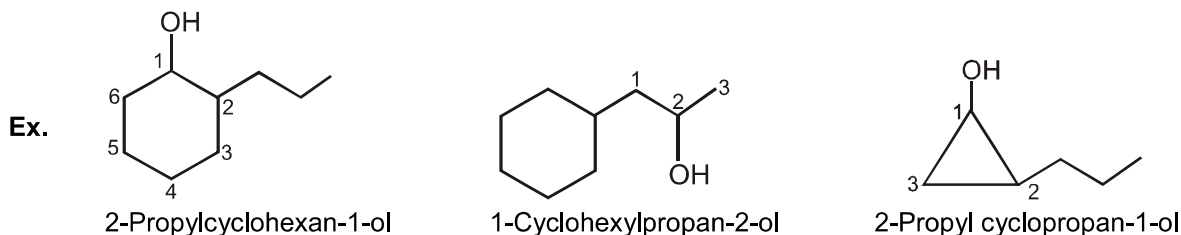
- (7) If a multiple bond and some other substituents are present in the ring, the numbering is done in such a way that the multiple bond gets the lowest number



- (8) If a compound contains an alicyclic ring directly linked to the benzene ring. It is named as a derivative of benzene.



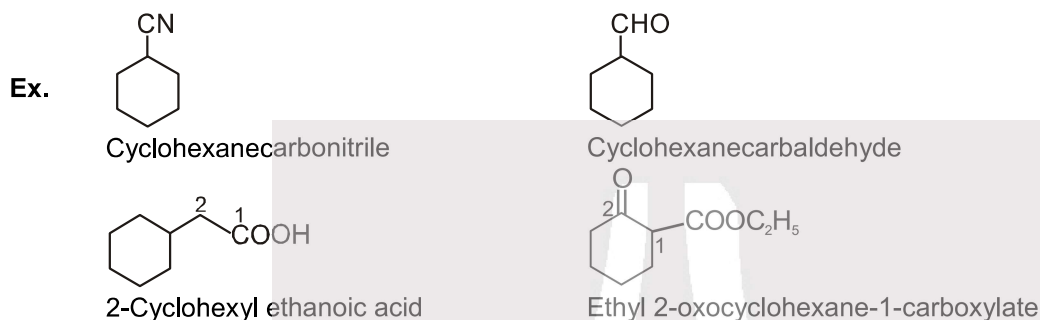
- (9) If functional group is present in cyclic compounds then the main chain is taken in which principal functional lie's, if the principal functional group is present in ring also then main chain will be taken for the maximum no. of carbon atoms.





- (10) When chain terminating functional group is directly attached with ring then ring is taken as parent chain & special suffix is used for this functional group.

| Functional Group  | Suffix            |
|-------------------|-------------------|
| CHO               | Carbaldehyde      |
| COOH              | Carboxylic Acid   |
| COX               | Carbonyl halide   |
| COOR              | Alkyl Carboxylate |
| CONH <sub>2</sub> | Carboxamide       |
| CN                | Carbonitrile      |



## 6. SECTION (D) : IUPAC NOMENCLATURE OF NON-CHAIN TERMINATING FUNCTIONAL GROUPS

Th10:

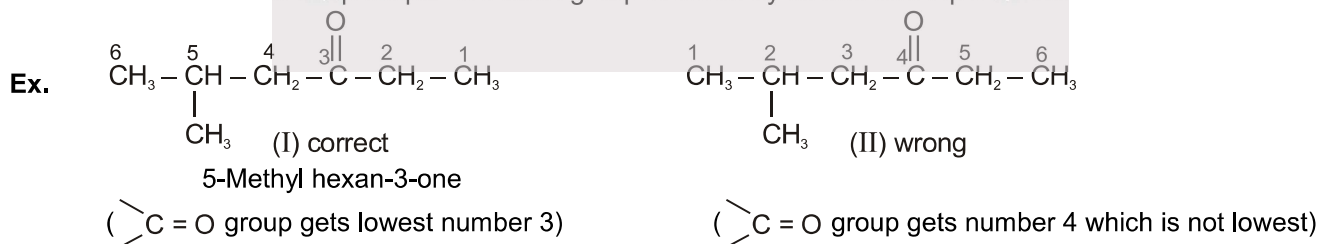
### 6.1 IUPAC nomenclature of compounds containing functional groups

#### 6.1.1 Rules for non chain terminating functional groups

- (1) **Parent chain** : Select the longest possible chain with maximum functional group and maximum unsaturation without caring whether it also denotes the longest possible chain or not.



- (2) **Lowest number for the functional group** : Numbering is done from that side of the chain which gives lowest locant to the principle functional group followed by double and triple bonds.



- (3) If a compound contains two or more like groups, the numerical prefixes di, tri, tetra etc. are used

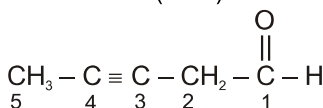
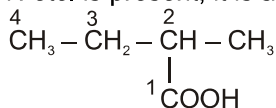




## 7. SECTION (E) : IUPAC NOMENCLATURE OF CHAIN TERMINATING FUNCTIONAL GROUPS

### 7.1 Rules for chain terminating functional groups

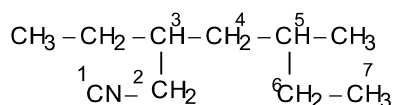
- (1) When a chain terminating functional group such as  $-\text{CHO}$ ,  $-\text{COOH}$ ,  $-\text{COOR}$ ,  $-\text{CONH}_2$ ,  $-\text{COCl}$ ,  $-\text{C}\equiv\text{N}$  etc. is present, it is always given number 1 (one.)



Ex. 2-Methylbutan-1-oic acid

Pent-3-yn-1-al

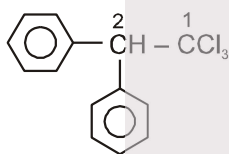
Que. Write the IUPAC name of



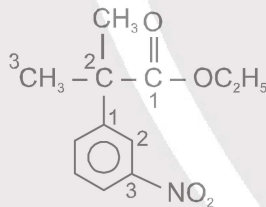
Sol.

- The longest chain containing functional group is of 7 carbon atoms. Therefore, the word root is hept & the chain is numbered as shown.
- There is no multiple bond in it. Hence, the primary suffix is **ane**.
- The functional groups is  $-\text{CN}$ . Hence, secondary suffix is **nitrile**.
- Moreover, there is a methyl group on carbon 5 and ethyl group on carbon 3.
- The IUPAC name is, therefore, **3-Ethyl-5-methylheptanenitrile**

- (2) The name for benzene as substituent is phenyl. In case the phenyl ring is further substituted, the carbon atoms of the ring directly attached to the parent chain in such a ways that the substituent on the ring gets the least possible number. For example

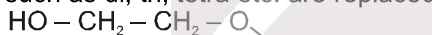


1,1,1-Trichloro-2,2-diphenylethane

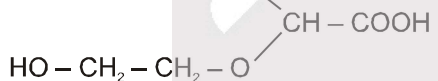


Ethyl- 2-methyl-2-(3-nitrophenyl) propanoate

- (3) If the organic molecule contains more than one similar complex substituents, then the numeral prefixes such as di, tri, tetra etc. are replaced by bis, tris, tetrakis etc. respectively.

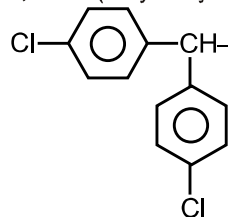


Ex.



2, 2-Bis (2-hydroxyethoxy) ethanoic acid

Ex.



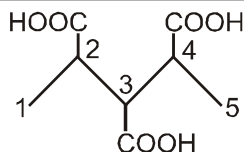
IUPAC Name : 1,1,1-Trichloro-2,2-bis(4-chlorophenyl)ethane

Common name is D.D.T. (Dichloro diphenyl trichloro ethane) & is used as insecticide.

- (4) When 3 or more principle functional groups are directly attached with an open chain, then special suffix is used.

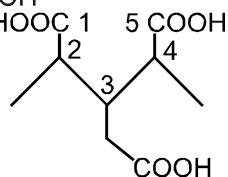


Ex.



Pentane- 2,3,4-tricarboxylic acid.

Not true for

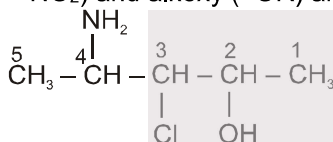


3-Carboxymethyl-2,4- dimethylpentanedioic acid

## 7.2 Rules for IUPAC nomenclature of polyfunctional compounds :

- (1) When an organic compound contains two or more different functional groups then senior functional group is selected as the principal functional group while other functional groups are treated as substituents.
- (2) Some functional group such as all halo groups (fluoro, bromo, chloro, iodo), nitroso (NO) nitro ( $-\text{NO}_2$ ) and alkoxy ( $-\text{OR}$ ) are always treated as substituent groups.

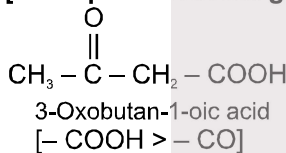
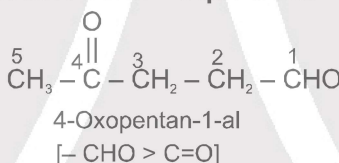
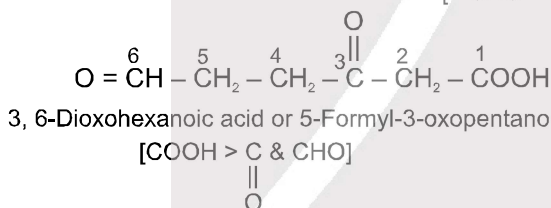
Ex.



4-Amino-3-chloropentan-2-ol

( $-\text{NH}_2$  &  $-\text{Cl}$  group treated as substituent)**Numbering the principal chain order is****[Principal functional group > double bond > triple bond > substituents]**

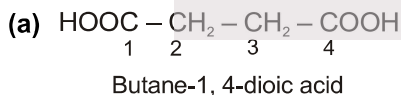
Ex.

3-Oxobutan-1-oic acid  
[ $-\text{COOH} > -\text{CO}$ ]4-Oxopentan-1-al  
[ $-\text{CHO} > \text{C}=\text{O}$ ]3, 6-Dioxohexanoic acid or 5-Formyl-3-oxopentanoic acid  
[ $\text{COOH} > \text{C} & \text{CHO}$ ]

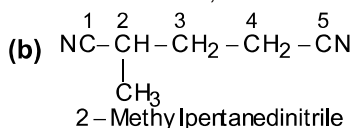
(3)

If more than one same chain terminating group are present then the principal chain is selected including the functional groups and numbring is done from that side which gives lowest locant to unsaturation and substituents.

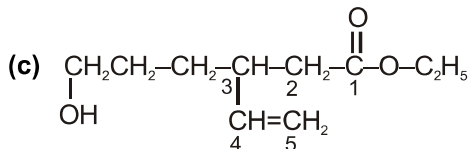
Ex.



Butane-1, 4-dioic acid



2-Methylpentanedinitrile



Ethyl-3-(3-hydroxypropyl) pent-4-enoate

Parent chain contains five rather than six carbon atoms.



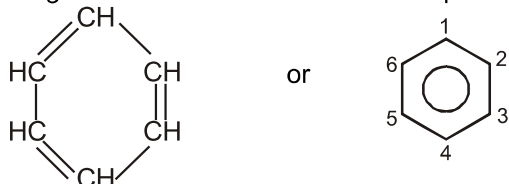


## 8. SECTION (F) : IUPAC-NOMENCLATURE OF AROMATIC COMPOUNDS

Th11:

### 8.1 Nomenclature of aromatic compounds

The aromatic compounds are cyclic compounds which contain one or more benzene type rings. Benzene is the simplest hydrocarbon of aromatic series which has planar cyclic ring of six carbon atoms having three double bonds in alternate positions as shown below.

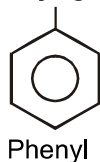


(i) **Nuclear substituted:** The functional group is directly attached to the benzene ring, in the IUPAC system they are named as derivatives of benzene. The position of the substituents in disubstituted benzenes are indicated either by prefixes such as o-(ortho) for 1,2, m-(meta) for 1, 3 and p-(para) for 1, 4 position. However, many of their common names have also been adopted by the IUPAC system.

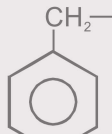
(ii) **Side chain substituted:** If functional group is present in the side chain of the benzene ring in the IUPAC system, these are usually named as phenyl derivatives of the corresponding aliphatic compounds.

The IUPAC and common names of a few important members of each family are given below.

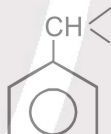
#### 1. Aryl groups :



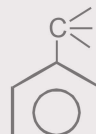
Phenyl



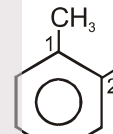
(Benzyl)



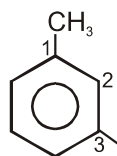
(Benzal)



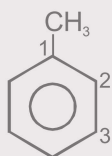
(Benzo)



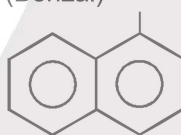
2-Tolyl or (o-Tolyl)



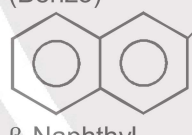
3-Tolyl  
or (m-Tolyl)



4-Tolyl  
or (p-Tolyl)



α-Naphthyl



β-Naphthyl

### 8.2 Other aromatic examples

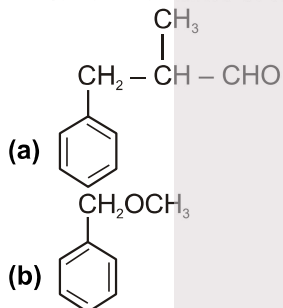
| S.No.                        | Compounds | Common Name    | IUPAC Name                 |
|------------------------------|-----------|----------------|----------------------------|
| <b>Aromatic Hydrocarbons</b> |           |                |                            |
| 1                            |           | Toluene        | Methylbenzene or Toluene   |
| 2                            |           | Xylene (o,m,p) | (o,m,p)<br>Dimethylbenzene |
| 3                            |           | Mesitylene     | 1,3,5-Trimethylbenzene     |





|   |  |              |                                       |
|---|--|--------------|---------------------------------------|
| 4 |  | Cumene       | Isopropylbenzene                      |
| 5 |  | Styrene      | Phenyl ethane<br>or<br>Ethenylbenzene |
| 6 |  | Naphthalene  | Naphthalene                           |
| 7 |  | Anthracene   | Anthracene                            |
| 8 |  | Phenanthrene | Phenanthrene                          |
| 9 |  | Pyrene       | Pyrene                                |

**Que.** Write IUPAC name of following aromatic compounds



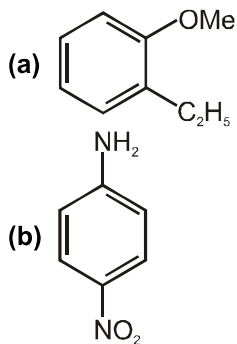
**Ans.**

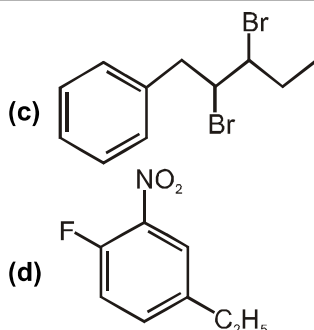
- (a) 2-Methyl-3-phenylpropanal  
(b) Methoxyphenylmethane (Benzyl methyl ether)

**Que.** Write the structural formula of :

- (a) o-Ethylanisole,  
(b) p-Nitroaniline,  
(c) 2,3-Dibromo-1-phenylpentane  
(d) 4-Ethyl-1-fluoro-2-nitrobenzene.

**Ans.**

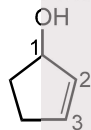




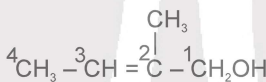
Th12:

## 9. SOME IMPORTANT 1993 RECOMMENDATIONS FOR IUPAC NOMENCLATURE OF ORGANIC COMPOUNDS :

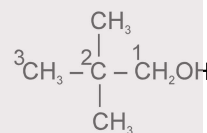
1. Locants (numerals and / or letters) are placed immediately before the part of the name to which they relate. For example :  
 $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$  should be named as but-1-ene ;  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$  should be named as propan-1-ol similarly, a few more examples are given as following :



Cyclopent-2-en-1-ol



2-Methylbut-2-en-1-ol



2,2-Dimethylpropan-1-ol

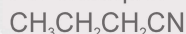
2. The locant 1 is often omitted when there is no ambiguity. For example.



Butanoic acid

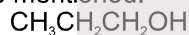


Propanal

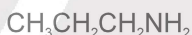


Butanenitrile

In all the above examples locant 1 for the functional group is omitted because the position of the functional group is unambiguous. However, in the following cases the position of the functional group must be mentioned.



Propan-1-ol

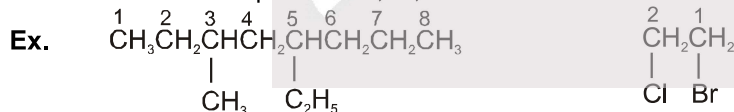


Propan-1-amine

Here, we cannot write simply propanol (or propanamine) because there are two propanols ; propan-1-ol and propan-2-ol.

3. Arrangement of Prefixes

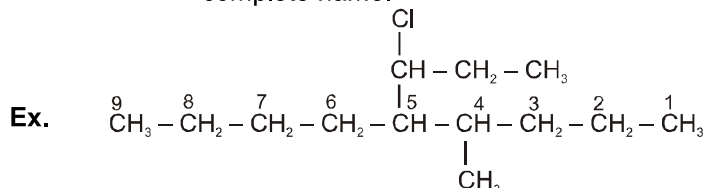
- (i) Simple prefixes such as methyl, ethyl, chloro, nitro, hydroxy, etc. are arranged alphabetically. The prefixes di, tri, etc. are however not considered for comparison.



5-Ethyl-3-methyloctane

1-Bromo-2-chloroethane

- (ii) The name of a prefix for a substituted substituent is considered to begin with the first letter of its complete name.



5-(1-Chloropropyl)-4-methyloctane

for the substituted 1-chloropropyl, 'C' is taken as the first letter.

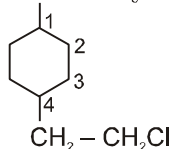




- (iii) When two or more prefixes consist of identical roman letters priority for citation is given to the group which contains the lowest locant at the first point of difference.



For example,



1-(1-Chloroethyl)-4-(2-chloroethyl)cyclohexane

Here, 1-chloroethyl gets priority over 2-chloroethyl.

## Structural Isomerism

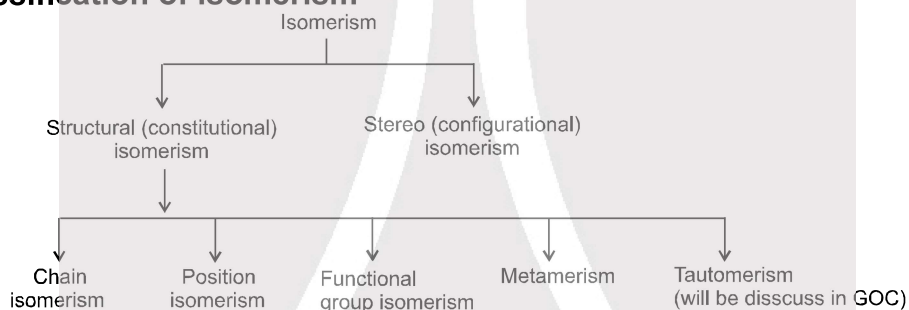
### 10. SECTION (G) : STRUCTURAL ISOMERISM

#### D4: Isomerism :

The phenomenon of existence of 2 or more compounds possessing the same molecular formula but different properties is known as isomerism. Such compounds are known as isomers.

Th13:

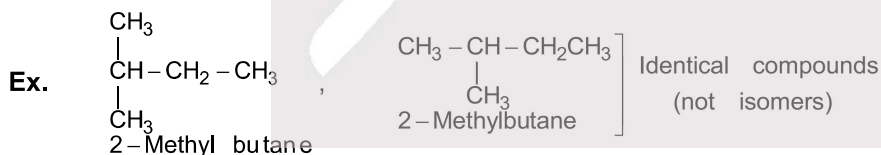
#### 10.1 Classification of isomerism



#### 10.2 Structural isomerism :

When two or more organic compounds have same molecular formula but different structural formula, (i.e., they differ in connectivity of atoms) then they are called **structural isomers** and the phenomenon is called structural isomerism.

Structural isomers have always different IUPAC name

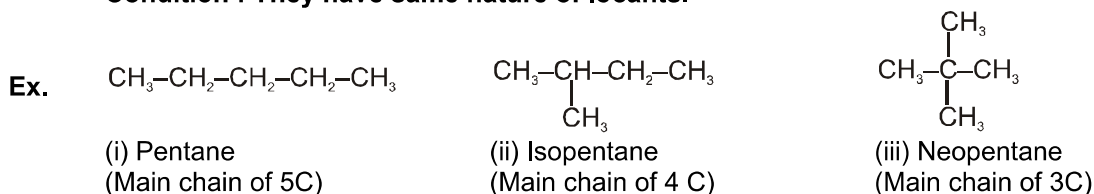


#### 10.3 Various types of structural isomers are :

D5:

- (a) **Chain isomerism:** Compounds having same molecular formula but different carbon skeletons (either difference in main chain or side chain) are known as chain isomers & phenomenon is known as chain isomerism.

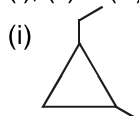
**Condition : They have same nature of locants.**





(i), (ii) & (iii) are chain isomers :

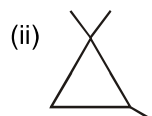
Ex.



Size of main chain = 3

Size of longest Side chain = 2

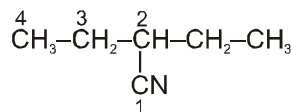
Both are chain isomers due to difference in number of carbon atoms in side chain.



Size of main chain = 3

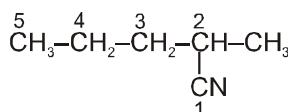
Size of longest side chain = 1

Ex.



2-Ethylbutanenitrile

&



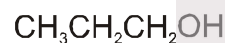
2-Methylpentanenitrile

D6:

(b)

**Position isomerism** : Compounds have same size of main chain & side chain along with same nature of locants but having different position of locants are known as position isomers & phenomenon is position isomerism.

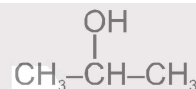
Ex.



Propan-1-ol

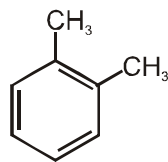
Difference only in position of -OH group

&



Propan-2-ol

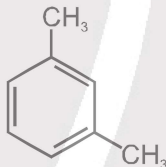
Ex.



o-Xylene

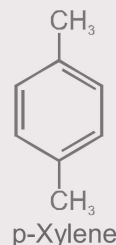
Difference only in position of -CH<sub>3</sub> group

&



m-Xylene

&



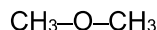
p-Xylene

D7:

(c)

**Functional isomerism** : Compounds having same molecular formula but different functional group are known as functional isomers & phenomenon is functional isomerism.

Ex.



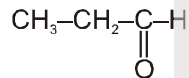
Methoxymethane

&



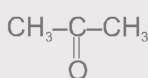
Ethanol

Ex.



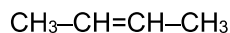
Propanal

&



Propanone

Ex.



But-2-ene

&

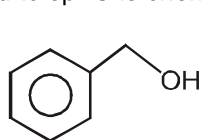


Cyclobutane

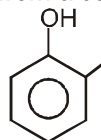
It is also known as **ring chain isomerism** [as 1 isomer has ring & another has aliphatic chain]

Note :

- (1) 1°, 2°, 3° amines are functional isomers.
- (2) 1°, 2°, 3° amides are functional isomers.
- (3) Alcohol attached to sp<sup>2</sup> C is chemically different from alcohol attached to sp<sup>3</sup> C.



&



are functional isomers.





- (4) Following compounds don't exist at room temperature therefore not considered as a structural isomer.

|     |                                                                       |      |                              |       |                                                                    |      |                                                                    |
|-----|-----------------------------------------------------------------------|------|------------------------------|-------|--------------------------------------------------------------------|------|--------------------------------------------------------------------|
| (i) | $\begin{array}{c} \text{—C=C—OH} \\   \quad   \end{array}$            | (ii) | $\text{—C}\equiv\text{C—OH}$ | (iii) | $\begin{array}{c}   \\ \text{—C—OH} \\   \\ \text{OH} \end{array}$ | (iv) | $\begin{array}{c}   \\ \text{—C—OH} \\   \\ \text{OR} \end{array}$ |
| (v) | $\begin{array}{c}   \\ \text{—C—O—C=C} \\   \\ \text{OH} \end{array}$ | (vi) | Any compound peroxy          | (vii) | $\begin{array}{c} \text{—C=C—} \\   \\ \text{NH}_2 \end{array}$    |      |                                                                    |

### D8:

- (d) **Metamerism** : Compounds having same nature of functional groups but different nature of alkyl groups along that polyvalent functional group are known as metamers & phenomenon is metamerism.

Ex.  $\text{CH}_3\text{—CH}_2\text{—O—CH}_2\text{—CH}_3$  &  $\text{CH}_3\text{—O—CH}_2\text{—CH}_2\text{—CH}_3$

Ethoxy ethane

Methoxy propane

[Ethyl groups on either sides of O.]

[Methyl & propyl groups on either sides of O.]

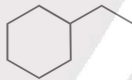
Ex.  $\text{CH}_3\text{—C(=O)—O—CH}_2\text{—CH}_3$  &  $\text{CH}_3\text{—CH}_2\text{—C(=O)—OCH}_3$  &  $\text{H—C(=O)—O—CH}_2\text{—CH}_2\text{—CH}_3$  are metamers.

Ex.  $\text{CH}_3\text{—CH}_2\text{—N(CH}_3\text{)—CH}_2\text{—CH}_2\text{—CH}_3$  &  $\text{CH}_3\text{—CH}_2\text{—N(CH}_3\text{)—CH}_2\text{—CH}_3$  are metamers.

Ex. Identify relationship between the given compounds :

(a) (i)  $\text{CH}_3\text{—CH}_2\text{CH}_2\text{—CH}_3$   
Butane  
Size of main chain = 4  
Size of side chain = 0  
Structure (i) & (ii) are chain isomers.

(ii)  $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{—CH—CH}_3 \end{array}$   
2-Methylpropane  
Size of main chain = 3  
Size of side chain = 1

(b) (i)   
1-Ethylcyclohexane  
Size of main chain = 6  
Size of side chain = 2  
Structure (i) & (ii) are chain isomers.

(ii)   
1,4-Dimethylcyclohexane  
Size of main chain = 6  
Size of side chain 1 = 1  
Size of side chain 2 = 1

(c) (i)   
Cyclohexane  
Size of main chain = 6  
Size of side chain = 0

(ii)   
1,2,3-Trimethylcyclopropane  
Size of main chain = 3  
Size of side chain 1 = 1  
Size of side chain 2 = 1  
Size of side chain 3 = 1

Structure (i) & (ii) are chain isomers.

(d)  $\begin{array}{l} \text{H}_3\text{C—CH}_2\text{—CH=CH}_2 \quad (\text{but-1-ene}) \\ \text{H}_3\text{C—CH=CH—CH}_2 \quad (\text{but-2-ene}) \end{array}$  position isomers

(e)  $\begin{array}{l} \text{HC}\equiv\text{C—CH}_2\text{—CH}_2\text{—CH}_3 \quad (\text{pent-1-yne}) \\ \text{H}_3\text{C—C}\equiv\text{C—CH}_2\text{—CH}_3 \quad (\text{pent-2-yne}) \end{array}$  position isomers





- (f) (i)  $\text{CH}_3\text{—CH}_2\text{OH}$  (Ethanol)  
Functional groups  $\text{—OH}$   
Structure (i) & (ii) are functional isomers.
- (ii)  $\text{CH}_3\text{—O—CH}_3$  (Methoxymethane)  
Function groups  $\text{—O—}$
- (iii)  $\text{CH}_3\text{—}\overset{\text{O}}{\parallel}\text{C—OH}$   
Ethanoic acid  
Functional groups  $\text{—COOH}$   
Structure (iii) & (iv) are functional isomers.
- (iv)  $\text{H—}\overset{\text{O}}{\parallel}\text{C—OCH}_3$   
Methyl methanoate  
Functional groups  $\text{—}\overset{\text{O}}{\parallel}\text{C—O—}$
- (g) (i)  $\text{C}_2\text{H}_5\text{—O—C}_2\text{H}_5$  (Diethyl ether)  
Hydrocarbon groups  $\text{—C}_2\text{H}_5$ ,  $\text{—C}_2\text{H}_5$   
Structure (i) & (ii) are metamers.
- (ii)  $\text{C}_3\text{H}_7\text{—O—CH}_3$  (Methyl propyl ether)  
Hydrocarbon groups  $\text{—C}_3\text{H}_7$ ,  $\text{—CH}_3$

